

LATNIVAARA, Sweden

WMO: 020380

Lat: 67.2551N

Lon: 20.2704E

Elev: 453

StdP: 96.00

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.6	(c) -28.8	(d) -35.0	(e) 0.1	(f) -31.4	(g) -31.9	(h) 0.2	(i) -28.6	(j) 8.7	(k) -2.3	(l) 7.5	(m) -4.2	(n) 0.3	(o) 340	(p) 0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.1	(c) 23.2	(d) 15.0	(e) 21.2	(f) 14.2	(g) 19.3	(h) 13.4	(i) 16.7	(j) 21.3	(k) 15.4	(l) 19.4	(m) 14.2	(n) 17.7	(o) 3.0	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 14.9	(b) 11.2	(c) 17.8	(d) 13.7	(e) 10.3	(f) 16.8	(g) 12.6	(h) 9.6	(i) 15.6	(j) 48.4	(k) 21.4	(l) 44.6	(m) 19.5	(n) 41.3	(o) 17.7	(p) 20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.4	5.5	-34.9	25.7	3.0	2.5	-37.1	27.5	-38.8	28.9	-40.5	30.3	-42.6	32.1
			-35.0	17.8	2.9	1.3	-37.0	18.7	-38.8	19.5	-40.4	20.2	-42.5	21.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-14.2	-13.5	-8.7	-2.3	3.9	9.1	12.4	10.0	5.3	-1.9	-8.4	-11.1		
	DBStd	10.56	7.34	7.32	5.65	4.42	3.90	3.67	3.09	3.31	3.50	4.82	6.07	6.89		
	HDD10.0	4380	751	657	578	369	194	59	11	41	144	370	552	654		
	HDD18.3	7254	1009	890	837	619	448	277	184	257	390	628	802	913		
	CDD10.0	170	0	0	0	0	4	32	87	42	4	0	0	0		
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH23.3	53	0	0	0	0	0	12	37	5	0	0	0	0		
	CDH26.7	4	0	0	0	0	0	1	3	0	0	0	0	0		
(14) Wind	WSAvg	2.3	1.6	1.9	2.6	2.6	2.8	2.9	2.5	2.3	2.4	2.1	1.7	1.9		
(15) Precipitation	PrecAvg	554	31	26	22	24	45	59	100	71	52	46	41	33		
	PrecMax	683	59	49	47	46	87	124	158	123	110	89	85	61		
	PrecMin	401	3	6	10	2	8	16	42	26	22	20	10	8		
	PrecStd	93	14	12	10	13	23	26	33	29	26	19	20	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.6	5.9	12.4	22.0	25.1	26.7	24.0	17.4	10.3	5.5	4.0		
		MCWB	2.2	2.4	2.7	6.4	11.3	16.1	17.3	16.5	12.2	7.7	3.7	2.2		
	2%	DB	1.3	1.8	3.5	8.7	18.2	21.6	24.0	21.6	15.1	8.3	3.2	1.4		
		MCWB	0.1	0.3	1.1	4.2	10.6	12.4	15.8	15.3	10.9	6.4	2.4	0.0		
	5%	DB	-1.4	-0.6	1.8	6.6	14.7	19.4	22.0	19.4	13.2	6.7	1.0	-0.5		
		MCWB	-2.1	-1.5	-0.1	3.0	8.8	11.8	15.2	14.5	10.1	5.3	0.5	-1.0		
	10%	DB	-3.9	-2.6	0.2	4.9	11.5	17.0	19.8	17.2	11.6	5.0	-0.3	-1.7		
		MCWB	-4.4	-3.2	-1.4	1.9	6.7	11.0	14.1	13.1	9.4	3.9	-0.6	-2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.7	2.7	6.7	12.7	16.7	18.5	18.3	13.2	8.4	4.1	2.3		
		MCDB	4.0	4.6	5.4	12.1	19.5	23.5	23.4	22.4	15.4	9.3	5.0	3.8		
	2%	WB	0.2	0.3	1.3	4.7	11.0	14.1	17.2	16.1	11.8	6.7	2.3	0.2		
		MCDB	1.3	1.6	3.2	8.1	17.6	17.9	22.2	19.9	14.2	8.0	3.3	1.0		
	5%	WB	-2.0	-1.5	0.1	3.3	8.9	12.7	16.0	15.0	10.7	5.3	0.5	-0.9		
		MCDB	-1.5	-0.5	1.6	6.1	14.4	17.4	20.2	18.7	12.4	6.5	0.9	-0.5		
	10%	WB	-4.3	-3.2	-1.3	2.2	7.2	11.5	14.8	13.7	9.8	3.9	-0.6	-2.2		
		MCDB	-3.9	-2.6	0.1	4.7	10.9	16.1	18.7	16.6	11.2	4.8	-0.3	-1.7		
(35) Mean Daily Temperature Range	MDBR		9.3	10.3	11.2	10.1	9.6	10.5	11.1	11.1	9.5	7.9	7.8	8.8		
	5% DB	MCDBR	8.6	8.9	10.7	11.6	16.1	15.4	15.5	15.3	12.3	9.2	6.3	7.8		
		MCWBR	8.1	8.0	8.8	7.5	9.5	7.8	8.3	9.3	8.3	7.3	5.9	7.5		
	5% WB	MCDBR	8.5	8.6	10.1	10.4	14.9	12.5	12.6	13.5	9.8	8.4	6.4	7.7		
		MCWBR	8.1	7.9	8.5	6.8	9.0	6.9	7.5	8.6	7.0	6.8	6.1	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.173	0.236	0.272	0.301	0.320	0.324	0.332	0.318	0.288	0.260	N/A	N/A		
	taud		1.934	2.148	2.234	2.277	2.463	2.519	2.508	2.529	2.586	2.383	N/A	N/A		
	Ebn at Noon		190	608	783	856	881	885	864	841	791	592	N/A	0		
	Edh at Noon		11	47	76	97	91	89	87	77	56	38	N/A	0		
(44) All-Sky Solar Radiation	RadAvg		0.05	0.54	2.01	3.89	4.91	5.05	4.71	3.44	1.90	0.65	0.12	0.00		
	RadStd		0.00	0.06	0.11	0.18	0.44	0.40	0.47	0.30	0.25	0.10	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page